cost component above can be further segmented into a trading related delay cost and an opportunity related delay cost. This is shown as follows:

$$\text{Delay Cost} = S \cdot (P_0 - P_d) = \underbrace{\left(S - \sum s_j\right) \cdot (P_0 - P_d)}_{\text{Opportunity Related Delay}} + \underbrace{\left(\sum s_j\right)(P_0 - P_d)}_{\text{Trading Related Delay}}$$

Analysts may wish to include all unexecuted shares in the opportunity cost component as a full measure of missed profitability.

It is important to point out that in many cases the analysts will not have the exact decision price of the manager since portfolio managers tend to keep their decision prices and reasons for the investment to themselves. However, analysts know the time the order was released to the market. Hence, the expanded implementation shortfall would follow our formulation above where we only analyze costs during market activity, that is, from t_0 to t_n. This is:

$$\text{Market Activity IS} = \underbrace{\left(\sum s_j\right)\left(P_{avg} - P_0\right)}_{\text{Trading Related}} + \underbrace{\left(S - \sum s_j\right)(P_n - P_0)}_{\text{Opportunity Cost}} + fees$$

Implementation Shortfall Formulation

The different formulations of implementation shortfall discussed above are:

$$IS = S \cdot \left(P_{avg} - P_d\right) + fees \tag{3.5}$$

$$\text{Perold } IS = \sum s_j \cdot \left(P_{avg} - P_d\right) + \left(S - \sum s_j\right) \cdot (P_n - P_d) + fees \tag{3.6}$$

$$\text{Wagner } IS = S(P_0 - P_d) + \left(\sum s_j\right)\left(P_{avg} - P_0\right) + \left(S - \sum s_j\right)(P_n - P_0) + fees \tag{3.7}$$

$$\text{Mkt Act. } IS = \left(\sum s_j\right)\left(P_{avg} - P_0\right) + \left(S - \sum s_j\right)(P_n - P_0) + fees \tag{3.8}$$

Trading Cost/Arrival Cost

The trading cost component is measured as the difference between the average execution price and the price of the stock at the time the order was entered into the market (arrival price). It is the most important metric to evaluate broker, venue, trader, or algorithmic performance, because it quantifies the cost that is directly attributable to trading and these specific parties. It follows directly from the trading related cost component from the expanded implementation shortfall. The investment related and